BUSINESS REQUIREMENTS DOCUMENT

AI-POWERED SELF-SERVICE RESTAURANT PLATFORM

Version 2.0
Production Spec

Project Name	Scan, Order & AI Concierge Web Platform	**Target Stack**	Next.js, FastAPI, Supabase, gpt-4o-mini
Primary Models	GPT-4o-mini (Generator & Judge)	**Voice Layer**	WebAudio / WebRTC + Fast STT/TTS
Payment Gate	Stripe Connect (Direct Payouts)	**Document ID**	BRD-2026-V2

1. Executive Summary & Core Intent

Small restaurants face severe operating margin pressure, front-of-house staff shortages, and peak-hour ordering bottlenecks. This project delivers an instant-access, Web-based ordering and AI concierge solution requiring **zero app downloads or registration** for diners and **minimal technical configuration** for owners.

Diners scan a dynamic table QR code to view menus, request dish modifications, verify dietary restrictions, interact via text or streaming full-duplex voice with an AI Waiter, and complete payments via Apple Pay/Google Pay. The system operates on a highly secure dual-LLM pipeline using `gpt-4o-mini` for both primary conversational tool execution and instantaneous validation (LLM-as-a-Judge).

2. Target User Roles & Objectives

User Role	Primary Objective	Key Success Metric
Diner / Customer	Order food quickly, clarify ingredients/allergens, pay securely via mobile wallet.	Order submission time under 2 minutes; zero app install barrier.
Kitchen Staff	Receive clear, paid, structured orders in real time without taking manual notes.	Zero missed tickets; audio alert reliability > 99.9%.
Restaurant Owner	Automate order intake, eliminate FOH staffing costs, auto-ingest menus via AI.	Onboarding completed in < 15 mins via photo menu upload.
Platform Admin	Manage multi-tenant subscriptions, track platform fees, monitor model safety.	Seamless automated revenue splits via Stripe Connect.

3. System Architecture & Dual-LLM Guardrail Flow

```
[QR Scan: Table / Rest ID] ──> [Next.js PWA Client] │
┌──────────────────────────┴──────────────────────────┐ ▼ ▼ [Direct UI Cart Select]
[Conversational Agent (Text/Voice)] │ │ │ (Generates Action JSON / Text) │ │ │ ▼ │ [Layer
1: Code Verification] (0ms) │ (Check Item ID & Base DB Prices) │ │ │ ▼ │ [Layer 2: gpt-4o-
mini Judge] (Sub-200ms) │ (Validate Allergen Safety & Grounding) │ │
└──────────────────────────┬──────────────────────────┘ ▼ [Validated Order Payload] │ [Stripe Connect
Payment] │ ▼ [Supabase Realtime Engine] ──> [Kitchen KDS / POS Printer]
```

4. Comprehensive Functional Requirements

4.1 Client Application & QR Routing

- `FR-1.1` **Dynamic Table Routing:** Resolves URLs formatted as `app.domain.com/r/{rest_slug}/t/{table_id}?token={sig}`. Invalid or missing tokens trigger dynamic local session creation.
- `FR-1.2` **Performance Budget:** First Contentful Paint (FCP) must complete within 1.0s on standard 4G connections; total initial bundle weight under 1.2MB.
- `FR-1.3` **Menu UI & Modifiers:** Displays items grouped by category, showing live stock tags, high-resolution imagery, allergen flags, and structured modifier pickers.

4.2 AI Concierge, Voice Engine & Dual-LLM Judge

- `FR-2.1` **Primary LLM Execution:** Utilizes `gpt-4o-mini` with strict system prompts and defined function-calling payloads (e.g., `add_to_cart(item_id, modifier_ids)`).
- `FR-2.2` **Deterministic Code Pre-Check:** The backend intercepts all function calls to enforce PostgreSQL database price lookups. The primary LLM is forbidden from setting or modifying currency values.
- `FR-2.3` **LLM-as-a-Judge Guardrail:** Evaluates non-deterministic responses (e.g., dietary suitability claims) using a second isolated `gpt-4o-mini` instance operating at `temperature: 0`. Ensures zero off-menu hallucinations.
- `FR-2.4` **Voice Interaction Pipeline:** WebAudio Push-To-Talk / continuous streaming mode feeding low-latency STT, LLM execution, and TTS. Total turn-around audio latency capped under 800ms.

4.3 Order Processing & Kitchen Operations

- `FR-3.1` **Stripe Connect Payment:** Payments must be captured prior to ticket creation. Platform fees are split atomically at point of sale directly into the merchant's bank account.
- `FR-3.2` **Kitchen Display System (KDS):** Web-based real-time dashboard powered by Supabase Realtime / WebSockets, featuring persistent audio alarms for unacknowledged orders.
- `FR-3.3` **Stock Overrides ("86ing"):** Staff can mark any item out-of-stock with one tap. This state immediately updates the client UI and removes the item from the active RAG prompt context.

5. Non-Functional, Security & Operational Requirements

> **Security Critical Principle: Zero Financial Trust in Generative Outputs**
> Generative models are treated purely as interaction layers. All mathematical operations, tax calculations, tip computations, and sub-totals are computed strictly on the backend application server using immutable database prices.

Category	Specification Details
Prompt Injection Defense	System prompts strictly sandbox user input. System enforces rigid JSON schemas for tool calls; non-conforming outputs are automatically discarded by the API Gateway.
Table Fraud Prevention	Cart sessions are bound to signed QR tokens and verified via browser geolocation bounds (optional toggle per restaurant) to prevent remote prank orders.
High Availability & Offline	KDS maintains continuous WebSocket heartbeat. Upon disconnect, it switches to 5-second polling fallback to ensure uninterrupted kitchen order intake.

Category	Specification Details
Multi-Tenant Isolation	PostgreSQL Row Level Security (RLS) policies enforce strict isolation across restaurant entities at the database kernel level.

6. Implementation Roadmap

1. **Phase 1 (Core Order & Pay):** QR routing, Next.js PWA, Supabase schema, Stripe Connect integration, KDS web dashboard.
2. **Phase 2 (Text AI Concierge):** Grounded RAG implementation, tool calling for cart additions, code pre-validation layer.
3. **Phase 3 (LLM-as-a-Judge):** Synchronous validation pipeline using `gpt-4o-mini` for allergen and grounding verification.
4. **Phase 4 (Voice Streaming & Vision Onboarding):** WebAudio integration for voice concierge and multimodal GPT-4o camera menu parser for owners.